



Product and technical documentation

AI-assisted B2B quoting SaaS for building trades

Version 2 · 17 August 2026

FastAPI · React · Supabase · Ollama · Hermes Agent

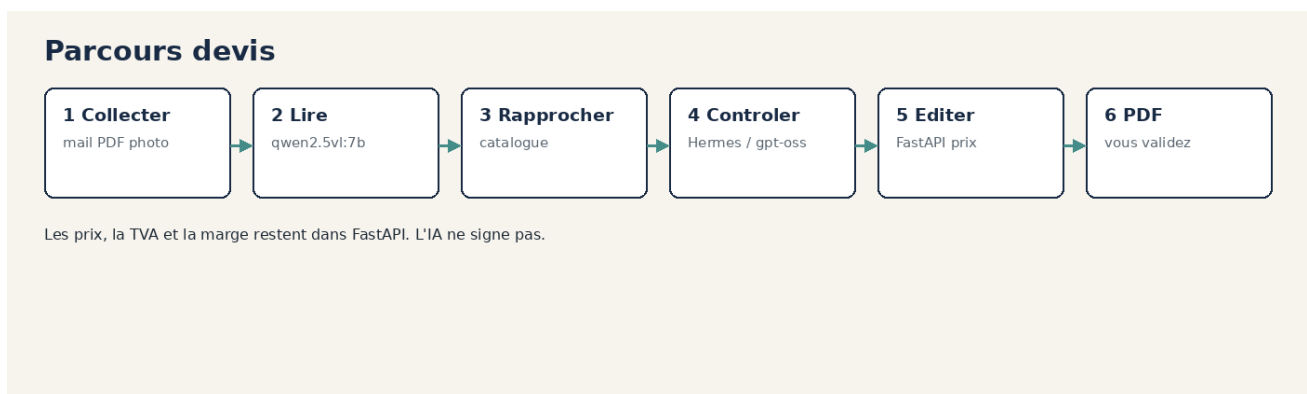
1. Overview

Blueseatra is a multi-tenant platform for SMEs in construction, multi-trade fit-out and maintenance. Each company has an isolated workspace: users, catalog, requests and quotes. The product reads a raw request (email, PDF, photo, text) and prepares a draft quote on **your** catalog prices. The AI never invents a tariff. You validate before sending.



Typical desk: the draft is ready, the last word stays human.

2. Quote path



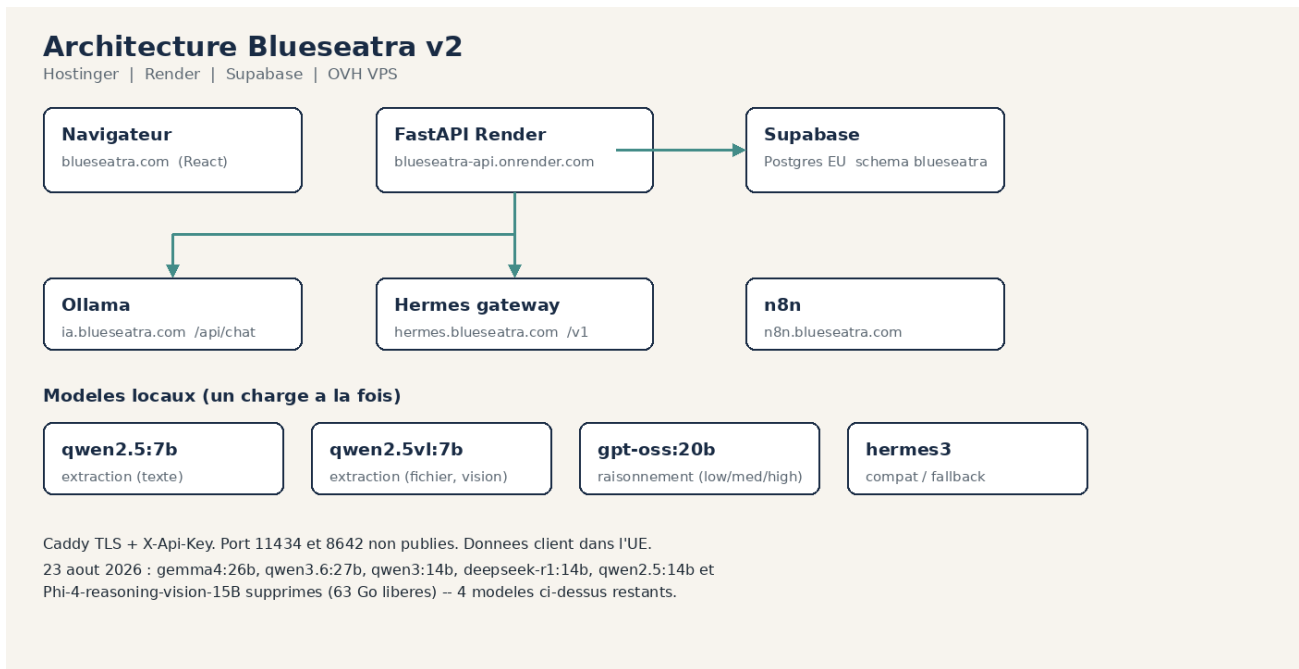
Collect, read, match, check, edit, PDF. Amounts (ex-VAT, French VAT 20/10/5.5/0, internal margin) stay in FastAPI. Hermes and Ollama do not sign.

3. Features

- JWT auth, roles owner / admin / operator / viewer / billing_admin, tenant_id isolation.
- Multilingual AI extraction (PDF, image, text) on qwen2.5:7b / qwen2.5vl:7b (vision).
- Material breakdown instead of repeating the title.
- Catalog matching with score and reason. Rematch when the catalog changes.
- TCE lots and sub-lots, labor hours, travel days, crew size.
- Editor: typed lines, per-line VAT, margin hidden on the client PDF, add from catalog.
- Pro forma PDF (SIRET, insurance, validity, acceptance).
- Audit log, members, FR/EN, product landing, lockup brand.

- Read-only MCP (requests, quotes, catalog) for the Perplexity agent.

4. Architecture v2



The frontend lives on Hostinger (manual React build). FastAPI runs on Render (Frankfurt). Postgres is on Supabase (EU). Local AI sits on an OVH Ubuntu 24.04 VPS (8 vCPU, 24 GB RAM, 200 GB NVMe, 162.19.44.2) with Ollama, Hermes Agent, n8n and Caddy.¹

Layer	Where	Role
React + Tailwind	Hostinger / blueseatra.com	UI, landing, editor
FastAPI	Render / blueseatra-api.onrender.com	Domain, prices, PDF, auth
PostgreSQL	Supabase eu-west-1	blueseatra schema
Ollama	ia.blueseatra.com	Local inference /api/chat
Hermes	hermes.blueseatra.com	Gateway /v1/chat/completions
n8n	n8n.blueseatra.com	Omnichannel intake (target)

Caddy terminates TLS and requires X-API-Key. Ports 11434 and 8642 are not published. API_SERVER_HOST=0.0.0.0 only on the Docker network.²

5. AI routing



Role	Model	Path
Extraction	qwen2.5:7b (text) / qwen2.5vl:7b (file, vision)	FastAPI → Ollama /api/chat
Materials reasoning	gpt-oss:20b, graduated think low/medium/high (default medium)	FastAPI → Ollama /api/chat direct
Reasoning fallback	hermes3 (no native reasoning, only remaining functional fallback)	Ollama
Price / VAT / margin	deterministic	FastAPI only

Updated 23 August 2026 (63 GB freed): gemma4:26b, qwen3.6:27b, qwen3:14b, deepseek-r1:14b, qwen2.5:14b and Phi-4-reasoning-vision-15B were removed from the VPS. gpt-oss:20b (already installed, 13 GB, 20.9B parameters, MXFP4 quantization) took over materials reasoning: the only remaining model with confirmed thinking capability (`ollama show gpt-oss:20b`), but it cannot disable reasoning -- only tune its depth via `HERMES_REASONING_EFFORT` (low/medium/high). No vision: file extraction stays on qwen2.5vl:7b. Hermes has no named reasoning slot: the main model is what the agent thinks with.⁴ `OLLAMA_MAX_LOADED_MODELS=1`. Ollama mem_limit 18g.

5b. As-is vs v2 map

Source: Cartographie Blueseatra v2 (16 August 2026). v1 = console + one LLM. v2 adds n8n omnichannel intake, Hermes multi-model routing, and one canonical request object. Card billing stays out of the first v2 cut.

Layer	As-is	v2 target
Intake	UI upload / paste	n8n: sites, IMAP, Drive, CRM, WhatsApp
Normalize	FastAPI requests.extracted	Canonical object in n8n before AI
AI	One Ollama call	14B extract, 27B writer, Gemma reasoner
Pricing	matching.py deterministic	Unchanged
Output	Editor + PDF	Same + email / CRM / Slack / ERP

Canonical object: tenant_id, source_type, source_ref, contact_*, message_text, attachments, site_address, requested_service, urgency, thread_id, received_at. Anti-replay statuses: received, analysed, to validate, quote generated.

Quote engine line types: material, labor, travel, note, page_break, lot, subplot. Auto TCE lots: structure, plaster, electrical, plumbing, HVAC, floors, paint, metalwork, maintenance. Skill rates: spot 0.45 h/unit, DHW cylinder 4.5 h, min visit 2 h, +1.25 h setup, max 7 h/day, labor rounded up. Default 42 EUR ex-VAT/h, 40 EUR/day Ile-de-France, supplies x 1.40.

5c. n8n P1 pack

Source: n8n P1 import guide. Workflows ship inactive, no secrets. Import one file at a time.

File	Role
WF-normalize-ai-quote.json	Hub Normalize then AI then Quote
WF-web-webhook.json	Client websites
WF-email-imap.json	Generic IMAP
WF-email-zoho.json	Zoho Mail imap.zoho.com:993
WF-email-gmail.json	Gmail
WF-email-outlook.json	Outlook 365

Credentials: Blueseatra JWT (owner Bearer) and one mailbox per tenant. Replace REPLACER_TENANT_ID. Activate the hub first, then one test channel. Hub: <https://n8n.blueseatra.com/webhook/blueseatra-normalize> Site: <https://n8n.blueseatra.com/webhook/blueseatra-intake> The pack does not compute prices and does not enable Drive/CRM/WhatsApp (P2+).

6. Data model

14 PostgreSQL tables, UUID keys, dynamic fields in JSONB. Every query filters on tenant_id.

Table	Role
users / tenants / tenant_users	Accounts, workspaces, roles
catalogs / catalog_versions / pricing_items	Versioned catalog, dynamic attributes
requests	Requests + extraction JSONB
quotes / quote_versions	Quotes, lines, price snapshot
import_jobs / import_errors	CSV imports
audit_logs	Audit
settings_integrations / company_profiles	AI, n8n, PDF legal block

7. Environment

Render backend (secrets stay out of Git):

Variable	Use
DATABASE_URL	Supabase pooler :6543
JWT_SECRET / APP_ENCRYPTION_KEY	Auth and secrets at rest
HERMES_BASE_URL	https://ia.blueseatra.com
HERMES_API_KEY	X-API-Key = VPS OLLAMA_API_KEY
HERMES_EXTRACT_MODEL	qwen2.5:7b
HERMES_REASONING_MODEL	gpt-oss:20b
HERMES_REASONING_EFFORT	medium (low/medium/high)
HERMES_GATEWAY_URL	https://hermes.blueseatra.com
HERMES_GATEWAY_KEY	VPS API_SERVER_KEY (Bearer)
MCP_API_KEY / MCP_TENANT_ID	Read-only connector
DB_SCHEMA	blueseatra (watch historical public in render.yaml)

VPS ovh-ai-stack: OLLAMA_API_KEY, API_SERVER_KEY, HERMES_MODEL=gpt-oss:20b. Never paste keys in chat.

8. Security

- Strong JWT, bcrypt, role re-read from the database.
- Tenant isolation and cross-tenant 404.
- Tenant AI keys encrypted with Fernet.
- Explicit CORS origins. 15 MB uploads.
- Caddy: TLS, X-API-Key, no model ports exposed.

- Hermes API: separate Bearer, max_concurrent_runs=1, 8642 unpublished.
- PostgREST public: RLS deny-all.

9. End-user path

- Create a workspace (/signup).
- Import the CSV catalog (map columns, activate a version).
- Create a request (text or file). AI extracts.
- Generate a draft. Check lines, lots, VAT.
- Add from catalog if needed (draft only).
- Fill the company profile. Validate. Download the PDF.



The catalog remains the only price source.

10. API (excerpt)

Method	Route	Role
POST	/api/auth/login	Login
POST	/api/requests	New request
POST	/api/requests/{id}/process	Re-run extraction
POST	/api/quotes/draft	Draft from a request
GET	/api/catalog/search	Catalog picker
GET	/api/quotes/{id}/pdf	Pro forma PDF
POST	/mcp	Read-only Perplexity tools

11. Frontend

Routes: / public landing, /login /signup, /app dashboard, requests, catalogs, quotes, members, audit, settings. Logo in public/brand/. Landing rewrite (PR 26): path, time saved, features, photos.



Identity: navy #193852, teal #438C88, paper #F7F4EE.

12. Deployment

- Frontend: git pull && yarn build then upload the build folder to Hostinger.
- Backend: Render auto-deploy from main (Frankfurt).
- AI: git pull on the VPS (github_ovh_ai_stack key) then docker compose up -d --force-recreate hermes.
- DNS A: ia / n8n / hermes.blueseatra.com → 162.19.44.2.

13. August 2026 log

Date	Deliverable
15-16 Aug	OVH stack, qwen3.6:27b, v2 map
17 Aug	gemma4:26b installed, ADR-003, Hermes API ADR-004
17 Aug	PR 22: extract 14B / reason Gemma via gateway
17 Aug	Quote test BS-2026-0017 (DHW cylinder)
17 Aug	Logo + landing (PRs 24, 25, 26)
23 Aug	Removed 6 VPS models (63 GB), repointed extract/vision
23 Aug	PR 47: X-Hermes-Session-Id (ADR-005 ovh-ai-stack)
23 Aug	PR 48: gpt-oss:20b reasoning model; PR 49: graduated level

14. Sources

1. Ollama gemma4:26b: <https://ollama.com/library/gemma4:26b>
2. Gemma + Ollama: <https://ai.google.dev/gemma/docs/integrations/ollama>
3. Hermes API server: <https://hermes-agent.nousresearch.com/docs/user-guide/features/api-server>
4. Hermes Docker: <https://hermes-agent.nousresearch.com/docs/user-guide/docker>
5. Hermes configuring models: <https://hermes-agent.nousresearch.com/docs/user-guide/configuring-models>